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Course: Fundamentals of data science

Course code: DSA0404

CO 3 - ASSESSMENT TOOL 2

Data Interpretation

6. Sampling Distribution Reasoning

Question:

A population has mean $\mu = 500$ and SD $\sigma = 80$. A random sample of $n = 64$ is drawn.

(a) Compute the standard error of the sample mean.

(b) What is the probability that the sample mean exceeds 520, assuming the sampling distribution is approximately normal?

(c) If n were increased to 256, how does the standard error change, and what principle explains this?

Answer:

Given:

$\mu = 500$,

$\sigma = 80$,

$n = 64$.

(a) Standard Error
$SE = \sigma / \sqrt{n}$
$SE = 80 / \sqrt{64} = 80 / 8 = 10$
Answer: Standard Error = 10.

(b) Probability that sample mean exceeds 520
$Z = (\bar{X} - \mu) / SE$
$Z = (520 - 500) / 10 = 2$

From the standard normal table: $P(Z > 2) = 0.0228$.
Answer: $P(\bar{X} > 520) = 0.0228$ or 2.28%.

(c) If $n = 256$
$SE = 80 / \sqrt{256} = 80 / 16 = 5$.
The standard error decreases from 10 to 5.

Principle: Standard error decreases as sample size increases: $SE \propto 1/\sqrt{n}$.

7. Covariance Calculation and Sign Interpretation

Question:

Given paired data on advertising spend (X, ₹'000) and sales (Y, ₹ lakh):

X: 10, 20, 30, 40, 50

Y: 15, 18, 25, 30, 42

Calculate Cov(X,Y). Is the relationship direction consistent with intuition? What does the magnitude alone fail to tell you (that correlation would)?

Answer:
Mean of X = (10 + 20 + 30 + 40 + 50) / 5 = 30.
Mean of Y = (15 + 18 + 25 + 30 + 42) / 5 = 26.

$\text{Cov}(X,Y) = \Sigma(X - \bar{X})(Y - \bar{Y}) / n$

Products:
$(-20)(-11) = 220$
$(-10)(-8) = 80$
$(0)(-1) = 0$
$(10)(4) = 40$
$(20)(16) = 320$

Sum = 660.

Cov(X,Y) = 660 / 5 = 132.

Answer:

Cov(X,Y) = 132.

Interpretation: Covariance is positive, indicating that advertising spend and sales tend to increase together. This is consistent with intuition.

Limitation: The magnitude of covariance alone does not indicate the standardized strength of the relationship because covariance depends on the units of measurement. Correlation is standardized and ranges from −1 to +1, making the strength and direction easier to interpret.

8. Correlation vs Causation Trap
Question:
A city reports that ice cream sales and drowning incidents have a correlation coefficient of r = 0.85. A newspaper claims "ice cream causes drowning." Identify the statistical flaw, name the likely confounding variable, and explain what additional evidence (not just correlation) would be needed to claim causation.

Answer:
Given

$r = 0.85$

This indicates a strong positive correlation between ice cream sales and drowning incidents.

Statistical flaw:
The newspaper makes the correlation-causation fallacy. Correlation does not prove that one variable causes another.

Likely confounding variable:
The likely confounding variable is hot weather/temperature. Hot weather can cause increased ice cream sales and more people to go swimming, which increases exposure to drowning risk.

Evidence required for causation:
1. Control for confounding variables.
2. Conduct appropriate controlled or experimental studies where feasible.
3. Use longitudinal evidence.
4. Establish a plausible causal mechanism.
5. Replicate the relationship across different settings.

Conclusion: A correlation of 0.85 does not prove causation.

9. Effect of Outliers on Correlation
Question:
A dataset of 9 points shows r = 0.42 (weak positive). One extreme outlier is removed, and r becomes 0.89. What does this reveal about the fragility of Pearson's correlation? Would Spearman's rank correlation be more robust here? Justify.

Answer:
Initially, $r = 0.42$

After removing the outlier,

$r = 0.89$.

The large change from 0.42 to 0.89 shows that Pearson's correlation is sensitive to extreme outliers. An outlier can strongly influence the covariance and standard deviations, causing the correlation coefficient to change significantly.

Spearman's correlation:
Yes, Spearman's rank correlation would generally be more robust. Spearman correlation uses the ranks of observations instead of their original numerical values. Therefore, an extreme numerical value has less influence on the result.

Comparison:
Pearson: Uses original values; sensitive to outliers; measures linear relationship.
Spearman: Uses ranks; more robust to outliers; measures monotonic relationship.

Conclusion: The example demonstrates that Pearson's correlation can be fragile when outliers are present.

10. Data Preparation — Missing Values
Question:
A dataset of 200 rows has 15% missing values in the "Income" column, and analysis shows these missing values are more common among unemployed respondents.
(a) Classify this missing data mechanism (MCAR/MAR/MNAR).
(b) Why would mean imputation introduce bias here?
(c) Suggest a better strategy.

Answer:
Given:

Total observations = 200

Missing Income = 15%

missingness is more common among unemployed respondents.

(a) Classification:
The missingness is related to employment status. If employment status is an observed variable in the dataset, the mechanism is MAR — Missing At Random.

(b) Why mean imputation causes bias:
Mean imputation replaces missing Income values with the overall average income. However, unemployed respondents are more likely to have missing Income, and their actual income may differ systematically from the overall average.

Therefore, mean imputation may:
• Distort the true income distribution.
• Reduce variability.
• Bias the estimated average.
• Affect relationships between Income and other variables.

(c) Better strategy:
A better strategy is multiple imputation or model-based imputation using relevant variables such as employment status, age, education, and occupation.

Conclusion: Instead of simple mean imputation, use multiple imputation/model-based imputation to obtain more reliable results.